

Cross-modal spatial gating of planning-MRI priors for longitudinal CBCT-to-CT synthesis

LaTIM · CMSG

MIART 2026 · MICCAI Workshop, Strasbourg

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Context: longitudinal CBCT-to-CT synthesis

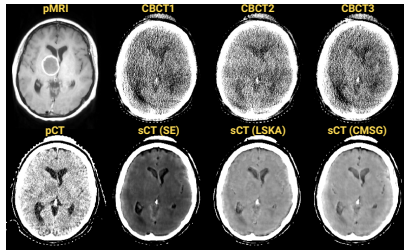
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- ▶ CBCT-to-CT synthesis exists — but **CBCT is not the only information available**.
- ▶ Across the course: **several CBCTs** per patient — an unexploited source of information.
- ▶ **Planning MRI** adds soft-tissue priors: Molina-Maza *et al.* (2025).



Planning MRI, three daily CBCTs, planning CT and the three synthesized CTs (soft-tissue window).

- ▶ **Our step further:** exploit the **multiple CBCTs** acquired during RT, with the planning MRI.
- ▶ **Ultimate objective:** update the **tumour visualization** from the **accumulation of CBCTs**.

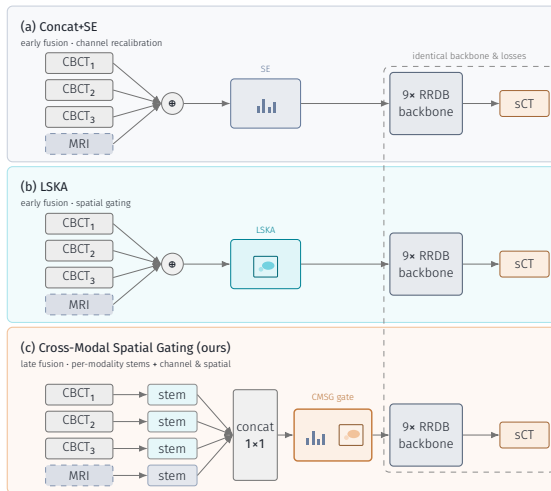
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Question

Does the *fusion mechanism itself* limit the benefit we can get from the planning-MRI prior?

Three fusion strategies



$\oplus$ = concat. (a) concat+SE (baseline); (b) LSKA spatial gate; (c) CMSG (ours), per-modality stems.

The baselines in detail: SE vs. LSKA

(a) Squeeze-and-Excitation - *channel*

- ▶ **Squeeze:** global average pooling $\Rightarrow$ one descriptor *per channel*.
- ▶ **Excitation:** 1×1 bottleneck ($\div 16$) + sigmoid $\Rightarrow \gamma \in \mathbb{R}^C$.
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(b) LSKA - *spatial*

- ▶ **Large separable kernels^a** (depthwise + dilated 1-D).
- ▶ **23×23 receptive field** at *linear* cost.
- ▶ **Per-pixel gate** $\Rightarrow$ suppresses local streaks, shading.
- ▶ **Why large?** streaks/shading span many pixels $\Rightarrow$ judge in **context**, not alone.

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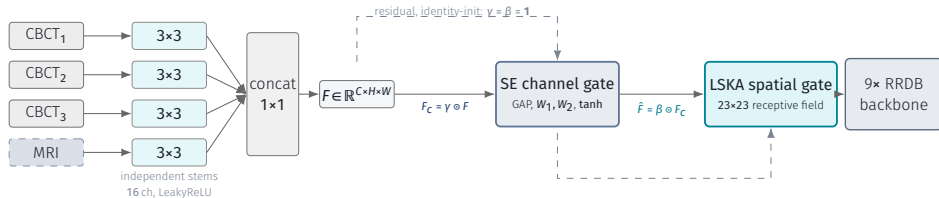
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Observations

SE: “*which channels?*”, a **global** weight. LSKA: “*where?*”, a **local** map.
For both SE and LSKA, **raw modalities** are concatenated at the input.

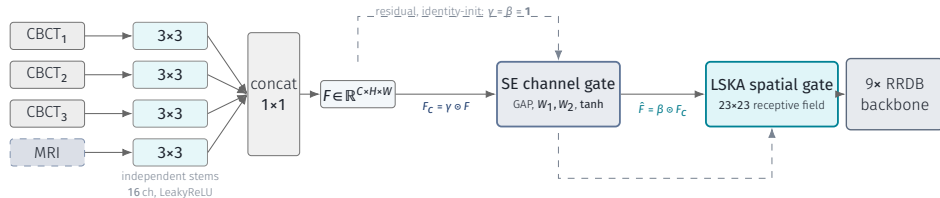
Proposed (c): cross-modal spatial gating (CMSG)



$$\gamma = 1 + \alpha \tanh(W_2 \text{ReLU}(W_1 \text{GAP}(F)))$$

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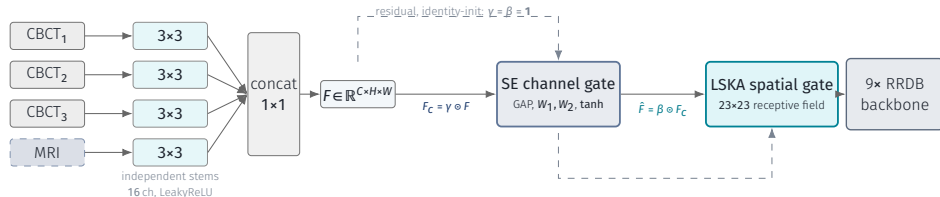
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- Late fusion: own 3x3 stem per input (16 ch) $\Rightarrow$ modality-specific features *before* mixing.

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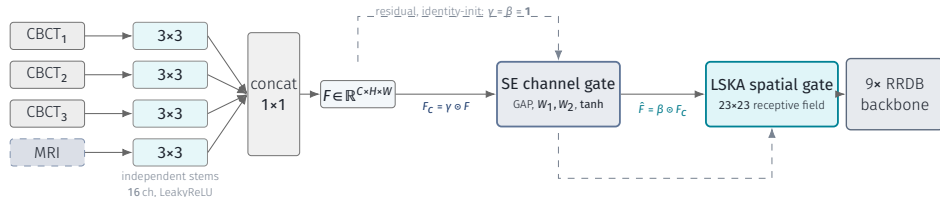
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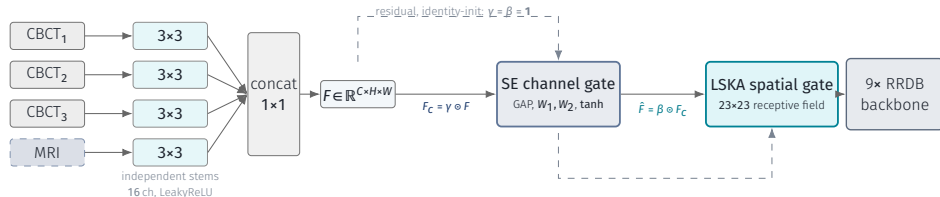
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γ reweights **channels** (SE), β reweights **pixels** (LSKA) - together they modulate the fused tensor.

Generator & discriminator

- ▶ **RRDB** cGAN: 9 blocks, growth **32, 64** channels;
- ▶ multi-scale **PatchGAN**.

Objective & training

- ▶ **L1/MS-SSIM + NEC** + adversarial;
- ▶ **15 epochs**, Adam 2×10^{-4} ; no tuning.

Shared backbone and losses

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Design principle

Identical backbone and losses across all strategies $\Rightarrow$ differences reflect the *fusion mechanism* only.

Cohort (single institution)

- ▶ **190** brain-metastasis patients, head mask;
- ▶ planning CT (GT), T1 **pMRI**, **3 daily CBCTs**;
- ▶ split **152 train / 38 hold-out** (patient-level).

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- ▶ body-mask n field-of-view;
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All images rigidly registered to the planning CT.

Results: ablation across inputs and fusion

	Concat+SE	LSKA	CMSG (ours)
<i>MAE (HU) ↓</i>			
1 CBCT	67.12 ± 12.87	67.65 ± 12.97	67.74 ± 12.92
1 CBCT+MRI	64.83 ± 10.70	62.64 ± 10.53	61.65 ± 11.20
3 CBCT	59.39 ± 11.57	59.81 ± 12.91	62.51 ± 10.81
3 CBCT+MRI	57.10 ± 11.73	58.19 ± 11.15	55.87 ± 10.53
<i>PSNR (dB) ↑</i>			
1 CBCT	29.23 ± 1.79	29.28 ± 1.73	29.19 ± 1.75
1 CBCT+MRI	29.99 ± 1.58	30.09 ± 1.58	30.05 ± 1.66
3 CBCT	30.34 ± 2.01	30.29 ± 2.10	30.23 ± 1.79
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≈ -7-8 HU.
- ▶ Best: 3CBCT + MRI ⇒ 30.98 dB,
55.87 HU.

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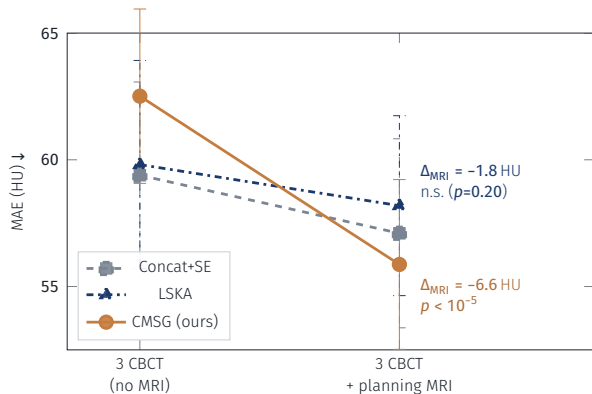
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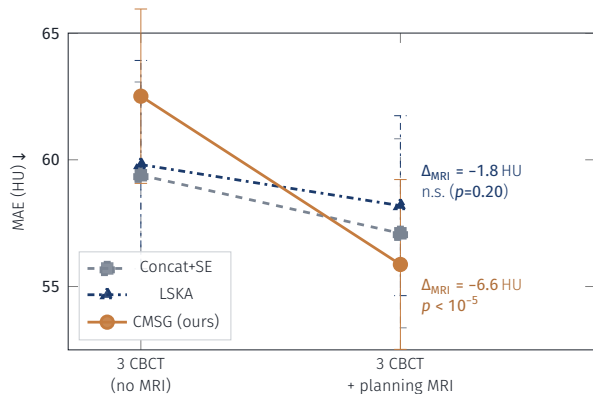
The MRI gain is **fusion-dependent**: under LSKA it barely helps.

Key finding: a fusion × modality interaction



Contrast (A vs. B)	Δ MAE (HU)	p
CMSG: 3CBCT+MRI vs. 3CBCT	-6.61	$< 10^{-5}$
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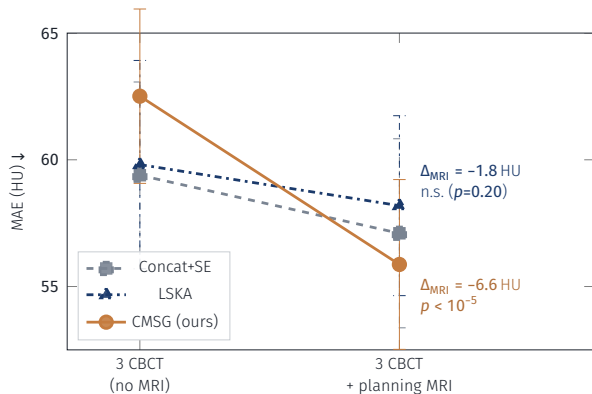
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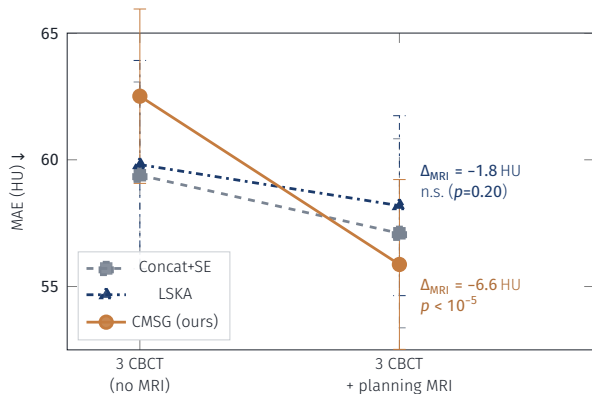
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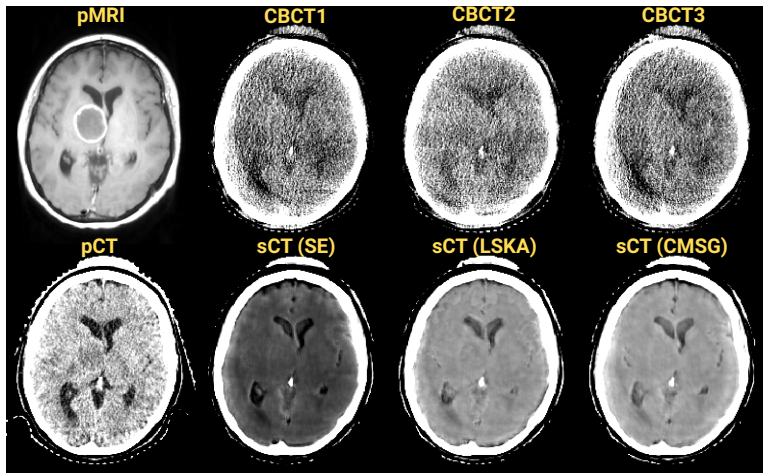
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- ▶ CMSG: -6.61 HU , $p < 10^{-5}$.
- ▶ LSKA: same MRI, not significant ($p = 0.20$).
- ▶ Net -3.06 HU vs. CBCT-only ($p = 2.2 \times 10^{-3}$).

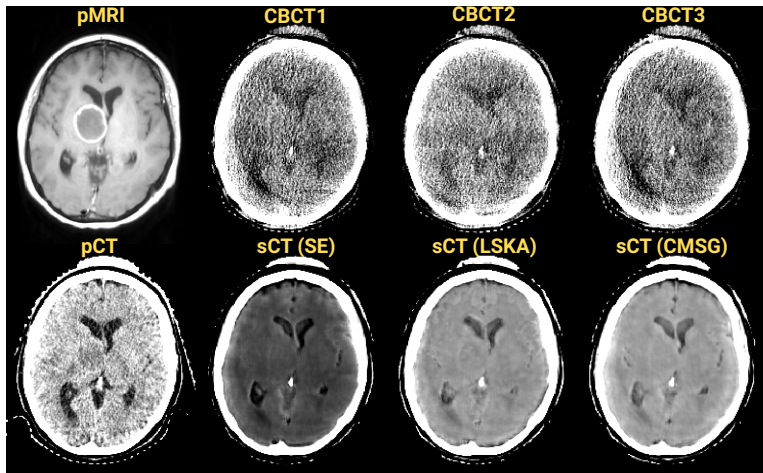
Qualitative results



What to see

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- ▶ CBCT geometry preserved (true on-couch anatomy).

💡 Insights

- ▶ The **fusion mechanism** - not MRI availability - governs the benefit.
- ▶ Own stems + channel & spatial gating ⇒ **significant** HU gain.
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⚠️ Limitations & next steps

- ▶ Single institution; no dosimetric validation yet.
- ▶ One training per configuration (seed variability).
- ▶ Next: **weight-shared, permutation-invariant** aggregation ⇒ update tumour visualization.

Thank you

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vincent.jaouen@imt-atlantique.fr · github.com/vhxjaouen/CrossModalSpatialGating



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Backup: paired contrasts (patient-level, $n = 38$)

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ΔMAE is the median of paired per-patient differences (A-B); negative favours model A. Conclusions are unchanged under Holm–Bonferroni correction.

References: X. Wang et al., ECCV-W 2018 (RRDB); J. Hu et al., CVPR 2018 (SE); K.-W. Lau et al., *Expert Syst. Appl.* 2024 (LSKA); V. Jaouen et al., ISBI 2024 (NEC); J. M. Molina-Maza et al., *J. Appl. Clin. Med. Phys.* 2025 (MRI + CBCT sCT).